Michael Murray

Senior Mechanical Design Engineer | Product Development | DFM, FEA, Manufacturing

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Portfolio & More: mjm-murray.github.io | linkedin.com/in/michaeljohnmurray

SUMMARY

Mechanical engineer with 17 years of end-to-end product development experience across medical devices, utility-scale solar tracker hardware, industrial power, automation, and more — from concept and prototyping through production. Holds three U.S. patents in electromechanical design. Deep hands-on expertise in DFM for machined, stamped, welded, and molded parts; FEA; GD&T; fastener engineering; and rapid prototyping (CNC, 3D printing, molding), with a record of delivering compliant, cost-effective hardware under standards from bodies like FDA, IEC, ISO, and ANSI. Leads cross-functional projects, qualifies suppliers, and mentors junior engineers.

PROFESSIONAL EXPERIENCE

Nevados Engineering | Oakland, CA

Senior Mechanical Design Engineer | April 2025 – October 2026

- Redesigned the stamped and welded tracker coupler assembly and took through production, improving structural performance and manufacturability; sized welds per AWS D1.1 and AISC 360.
- Performed linear and nonlinear FEA (SOLIDWORKS Simulation) on weldments, couplers, torque tubes, and bolted joints under wind downforce, uplift, and torsion to validate factors of safety.
- Led DFMEA of the coupler assembly and validated designs through stiffness, ultimate-strength, and fatigue. Conducted and analyzed lifecycle testing.
- Achieved consensus through high-level status meetings with the C-suite and major stakeholders.
- Owned fastener engineering and supplier quality: torque–tension and K-factor analysis (VDI 2230, ISO 16047), PPAP review, and supplier corrective actions; led fastener coating analysis and testing.
- Qualified gearmotor suppliers for slew drives, including torque-curve evaluation, splined-shaft interface review, and an output-shaft failure investigation.
- Created coating specs with suppliers for long-term outdoor exposure by site corrosivity; qualified via cyclic (IEC 60068-2-52) and continuous (ASTM B117) salt-spray testing, graded per ASTM D610.
- Standardized SOLIDWORKS and PDM workflows, improving consistency and team efficiency.

Swope Design Solutions | San Francisco, CA

Senior Mechanical Engineer | June 2022 – April 2025

- Led design and build of diaphragm pump-to-catheter system for removing arterial calcified lesions, from proof of concept through tested prototypes; served as customer-facing project manager.
- Designed and developed a positioning mechanism for a surgical procedure to treat kidney stones.
- Designed and built an automated data matrix scanning system, coding vision and analysis logic in C and Python. Motor-driven positioning system for automated movement.
- Designed a heater subsystem for a medical device, performing thermal analysis and benchtop testing.
- Created injection-molded part design and tooling, machined molds, and molded parts in silicone.
- Redesigned endoscopic robot components for injection molding, worked with overseas CM on moldability.
- Mentored junior engineers and ensured alignment with project milestones.

True Digital Surgery | Santa Barbara, CA

Senior Staff Mechanical Design Engineer | November 2020 – December 2021

- Designed mechanical subsystems of a positioning system for a robotic neurosurgical 3D microscope.
- Performed FEA to prevent mechanical failures and confirm factors of safety per IEC 60601 requirements.
- Performed thermal analysis to maintain mechanical integrity under heat application.
- Developed fit/function checks using GD&T in accordance with ASME Y14.5 standards.

Surround Medical Systems | Morrisville, NC

Senior Mechanical Engineer | May 2019 – September 2020

- Led the mechanical department in designing a 3D X-ray system from concept through production.
- Improved x-ray delivery to reduce patient radiation exposure while maintaining image quality.
- Co-authored a 510(k) submission for FDA clearance.
- Prevented mechanical failures through FEA and FMEA studies.

ABB | Raleigh, NC
Senior Mechanical Engineer, R&D | March 2017 – May 2019

- Standardized transformer designs using common parts, reducing cost and lead times.
- Enhanced mechanical designs of transformers/enclosures to increase convection efficiency.
- Led a project to automate creation of drawings, models, and data, writing a standard for programming.
- Developed an outdoor-rated transformer cabinet, incorporating weatherproof coating.

Ingersoll Rand | Davidson, NC
Design Engineer III | August 2014 – March 2017

- Designed parts and assemblies for complex, custom industrial centrifugal air compressors (2,000–7,000 CFM, 400–1,250 HP) to meet ANSI and ISO standards.
- Designed dual oil cooler system for 5,000-cfm air compressors.
- Optimized airflow through pre-build assessments, improving compressor performance.

Newell Rubbermaid – Piedmont Hardware Brands | Charlotte, NC
Mechanical Design Engineer | February 2013 – August 2014

- Designed high-volume sheet metal, die-cast, injection-molded, and stamped parts.
- Utilized hand calcs and SOLIDWORKS Simulation for FEA, guiding new product designs.
- Prevented failure modes through Computer-Aided Engineering (CAE).

Sandvik | Mebane, NC
Manufacturing Engineer | September 2011 – February 2013

- Created CAD models, CNC programs through CAM, and drawings for high-volume metal-cutting tools.
- Reduced tooling costs by 30% via optimized CAM programs for 5-axis mills, lathes, and grinders.
- Designed and implemented a coolant reclamation system to optimize machining efficiency and cut waste.

ABB | Lake Mary, FL
Mechanical Design Engineer | August 2009 – September 2011

- Invented a remotely operated device for industrial medium-voltage circuit breakers (5–15 kV); recognized with two patents (U.S. Patents 9,876,335 and 8,654,513B2).
- Developed a Modular Mechanism Operated Cell to retrofit circuit breakers (Patent 8,878,088).
- Improved circuit breaker thermal flow to prevent overheating and brazing issues.

TECHNICAL SKILLS

DFM, DFA, DFx: Machined, stamped, welded, cast, sheet metal, and molded parts

Mechanical Design, CAD & Drafting: SOLIDWORKS, Creo, NX, Inventor, Onshape, Fusion

Finite Element Analysis (FEA): SOLIDWORKS Simulation (static, linear, nonlinear, modal, thermal)

Geometric Dimensioning & Tolerancing (GD&T): ASME Y14.5, tolerance stack analysis

Product Development & Project Management: Jira, Confluence, Monday.com, and more

Rapid Prototyping & 3D Printing: FDM, SLA, SLS, DMLS; CNC & manual machining; molding

Supplier Quality, Management, and Identification: PPAP, FAI, SCAR, qualification, supplier audits

Failure Mode & Effects Analysis (FMEA / DFMEA)

PDM / PLM: SOLIDWORKS PDM Professional administration, Arena, Windchill

Regulations & Standards: FDA, IEC 60601, ISO 13485, ASME, ASTM, ANSI, AWS, AISC, UL, IEEE, more

Thermal Analysis: SOLIDWORKS Simulation, Flow Simulation

CAM (Computer-Aided Manufacturing): HSMWorks, NX

Injection Molding & Plastics

EDUCATION

North Carolina State University, B.S. Mechanical Engineering, 2009
Honors: cum laude, Dean's List, 2008 NCSU ME Design Award, Co-Op Program

ADDITIONAL NOTES

Portfolio, Manufacturing Process Experience, and more: mjm-murray.github.io